NAAMA SKITAL

SOFTWARE ENGINEER | PYTHON | SOFTWARE SYSTEMS

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PROFILE

Computer Science Honors graduate (GPA 92.17) with professional Python/Linux software development experience. Built diagnostics, validation workflows and distributed sensor messaging components, with strong fundamentals in OOP, data structures, algorithms and testing. Interested in backend engineering and reliable software systems.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL **Core:** Algorithms, Data Structures, OOP, Debugging, Testing
AI / Data: PyTorch, scikit-learn, NumPy, pandas, CLIP, NLP, Computer Vision, Model Evaluation
Systems / Tools: Linux, Git, ROS2, DDS, Raspberry Pi, Sensor Synchronization, Jupyter

PROFESSIONAL EXPERIENCE

Software Engineer | LMY *2025–2026*

- Developed Python software for a Linux-based UAV payload collecting LiDAR and navigation sensor data for 3D mapping workflows.
- Worked with ROS2/DDS messaging, sensor timing and synchronization, Raspberry Pi services, and data acquisition pipelines.
- Built validation and diagnostic scripts; investigated asynchronous data, communication delays, and timing inconsistencies.
- Supported post-flight processing and troubleshooting across embedded and ground-side software components.

SELECTED PROJECTS

Multimodal Advertisement Moderation | Python, PyTorch, CLIP, OCR GitHub

- Built and evaluated a four-class image moderation pipeline; compared frozen CLIP, partial fine-tuning, and image/OCR fusion.
- Best fusion model achieved **91.00% accuracy** and **0.6551 macro-F1**; analyzed imbalance and calibration.

Context-Aware Sarcasm Detection | NLP, MiniLM, scikit-learn GitHub

- Compared TF-IDF/logistic regression and sentence-embedding baselines across comment-only and context-aware inputs.
- Achieved **72.53% accuracy** with the comment-only TF-IDF baseline; examined the effect of context.

Topological Data Analysis | Python, GUDHI, Persistent Homology GitHub

- Studied Vietoris–Rips filtrations, H_0/H_1 persistence, diagrams and stability under perturbations on synthetic point clouds.

Software and Algorithms | C++, Java C++ / Java

- Built nine C++ coursework implementations covering OOP, templates, STL and generic containers; implemented Java Bayesian network inference with variable elimination.

Java Bayesian Network Inference Engine | Java, OOP, Algorithms GitHub

- Implemented XML-based Bayesian network parsing, probabilistic queries and variable elimination with operation counting.

EDUCATION

B.Sc. Computer Science | Ariel University *Nov 2023 – Aug 2026*
Honors Program (AI / Data Science) · GPA: **92.17** · Rector's Excellence (first year)
Relevant coursework: Algorithms, Data Structures, Operating Systems, Computer Networks, ML, Deep Learning, NLP and Computer Vision.

NATIONAL SERVICE, TEACHING & VOLUNTEERING

National Service (2 years): Supported at-risk youth at Dolev residential school and a Jerusalem high school for students who had left mainstream education; taught mathematics.
Mathematics Teaching: Prepared at-risk high-school students for the 4-unit mathematics matriculation examination.
Community Volunteering: Bnei Akiva youth counselor, including mentoring children with special needs; volunteered with children at Schneider Children's Medical Center.